

1 Formula Management

Overview

HYDRA menu	System administration --> System administration --> Formula management
FEDRA menu	System administration --> System administration --> Formula management
Transaction code	foma
Function authorization	mdfoma

Purpose

Formulas are defined for different purposes within the formula management function, e.g. to convert quantities into alternative units or even to determine process times.

Integration

The formula management function is used by different components within the system. These components are, among others:

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- MES Link & Enabling (MLE): to calculate default values and/or to determine conditions
- Material & Production Logistics (MPL): Conversion of quantities into other quantity units
- Shop Floor Data Collection (BDE): To calculate the processing time
- Shop Floor Data Collection (BDE): to convert quantities into other quantity units
- ←**Selection criteria**

The application provides the following selection criteria:

formula

Formula name

Type

Restricts the formula types

Field descriptions

formula

Formula key;

Unique within the context, e.g. 0001

→Type

Type of formula to differentiate logically:

- | | |
|-----|---|
| 1 | Quantity conversion |
| 2 | Determination of process time |
| 3 | Determination of remaining run time |
| 7 | Formulas to restrict composition |
| 8 | Formulas for the composition scrap rate |
| 50 | MLE (MES Link&Enabling) conditions |
| 999 | User-defined |
| 60 | Workplace formula (MES-Cockpit) |
| 61 | Order formula (MES-Cockpit) |

- | | |
|----|---------------------------------|
| 62 | Operation formula (MES-Cockpit) |
| 63 | General formulas (MES-Cockpit) |

The selected type is decisive for the transfer of formulas and KPIs into MES-Cockpit. As formulas are filtered based on the selected type.

Responsibility area

Assignment of a responsibility area

Comment Designation or description of the formula

Editable

This value is set for the formulas delivered by MPDV; formulas that are configured with "editable = N" can neither be changed nor deleted.

Functions, operators and absolute terms to be used in formulas and conditions

You can use the following operators and expressions to formulate conditions:

Functions	
abs(x)	Calculates the absolute value
atan(x)	Calculates the arc tangent
cosh(x)	Calculates the hyperbolic cosine
float(x)	Converts the value into a floating point number
sqrt(x)	Calculates the square root
acos(x)	Calculates the arc cosine
atan2(y,x)	Calculates the arc tangent of y/x
exp(x)	Calculates the exponential value
log(x)	Calculates the logarithm
sin(x)	Calculates the sine
tan(x)	Calculates the tangent
asin(x)	Calculates the arc sine
cos(x)	Calculates the cosine
int(x)	Converts the value into an integer
log10(x)	Calculates the logarithm
round(x)	Rounds to integer value
round(x,y)	Rounds the value x to y decimal places
sinh(x)	Calculates the hyperbolic sine
tanh(x)	Calculates the tangens hyperbolicus
trunc(x)	Reduces the value x to an integer value
trunc(x,y)	Reduces the value x to y decimal places
string(x)	Converts the value into a character string

Operators	
x and y	The AND operator returns the value 1 (true) if both conditions are TRUE. Otherwise, the value 0 (false) is returned.
x or y	The OR operator returns the value 1 (true) if one of the conditions is TRUE. Otherwise, the value 0 (false) is returned.
x like y	The comparison operator "like" returns the value 1 (true) if the first operand matches the pattern of the second operand. Otherwise, the value 0 (false) is returned. The pattern may include the following placeholders: "*" --> 0 -n any characters "?"--> exactly one (random) character
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!x	The logical negation operator "NOT" returns the value 0 (false) if the value is unequal to 0 (true), and the value 1 (true) if the value is 0 (false).
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x + y	Addition
x – y	Subtraction
x / y	Division
x * y	Multiplication
x ** y	Calculates x to the power of y
x << y x >> y	The bitwise shift operators shift the first operand to the left (<<) or to the right (>>) by the number of positions specified by the second operand.
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x < y	Returns 1 (true) if the value x is less than y, otherwise 0 (false)
x <= y	Returns 1 (true) if the value x is less than or equal to y, otherwise 0 (false)
x > y	Returns 1 (true) if the value x is greater than y, otherwise 0 (false)
x >= y	Returns 1 (true) if the value x is greater than or equal to y, otherwise 0 (false)
x == y	Returns 1 (true) if the value x is equal to y, otherwise 0 (false)
x != y	Returns 1 (true) if the value x is unequal to y, otherwise 0 (false)

$x \& y$	The bitwise AND operator compares every bit of the first operand with the corresponding bit of the second operand. If both bits are 1 the bit is set to 1 in the resulting value. Otherwise, the bit is set to 0 in the resulting value.
$x \wedge y$	The bitwise Exclusive-OR operator compares each bit of the first operand with the corresponding bit of the second operand. If one bit is 0 and the other bit is 1 the bit is set to 1 in the resulting value. Otherwise, the bit in the resulting value is set to 0.
$x y$	The bitwise Inclusive-OR operator compares each bit of the first operand with the corresponding bit of the second operand. If one bit is 1 the bit is set to 1 in the resulting value. Otherwise, the bit is set to 0 in the resulting value.

Constants	
pi	3.141592654
e	2.718281828

Distinctive features of the formula definition for the MES cockpit (applies only when using HYDRA).

Formula definitions used for MES-Cockpit provide the following features:

- Linking formulas to each other

With the specification @(<Formula2>) a calculation can be based on an existing formula definition

- Definition of conditions

Field contents can be checked by defining conditions. Example: `if(w.cycle.target > 0, w.rpa11, 0)`

→ The duration posted on the resource performance account should only be integrated in the calculation if the target cycle is greater than 0.

Deployment in the BDE (only relevant if HYDRA is in use)

Formulas can be used to calculate the following values (provided that they are not collected explicitly):

- Waiting time
- Setup time
- Processing time
- Inspection time
- Teardown/retooling time
- Target cycle The target cycle has been designed to monitor machines based on cycles in the machine data collection (HYDRA-MDE).
- Remaining run time The remaining run time of an operation is calculated based on the formula that is defined for the operation (tab durations > RRT formula).
→→→→→

The corresponding formula fields of the operation refer to the respective formulas defined in the formula management function. In addition to the basic arithmetic operations, it can be referred to the operation fields listed below. These are fields that are defined as target values for the operation and fields that are available as status information (actual values).

Target values at the operation

Target quantity: base quantity unit	ANR.SGR:GUTB
Target quantity: primary quantity unit	ANR.SGR:GUTP / ANR.SGR:GUT
Target quantity: secondary quantity unit	ANR.SGR:GUTS
Target quantity: tertiary quantity unit	ANR.SGR:GUTT
Target scrap: base quantity unit	ANR.SGR:AUSB
Target scrap: primary quantity unit	ANR.SGR:AUSP / ANR.SGR:AUS
Target scrap: secondary quantity unit	ANR.SGR:AUSS
Target scrap: tertiary quantity unit	ANR.SGR:AUST
Unit quantity	ANR.EINHMENGE
Target cycle	ANR.SZY
Partitioning	ANR.TLG
Factor	ANR.IMPFAKT
Waiting time	ANR.WARTZ
Setup time	ANR.RUEZ
Additional setup time (due to setup change)	ANR.RUEZ:ZUSCHL
Processing time	ANR.BEARBZ
Inspection time	ANR.PZ

Target values at the operation

Teardown/retooling time	ANR.ABRZ
Delivery time	ANR.LIEZ
Transport time	ANR.TPZ
Delivery time (external production)	ANR.LIZ
Number of splits	ANR.ANZSPLIT
Default values 1 ... 10	ANR.VGW01 ... ANR.VGW10

Actual values at the operation (OP status)

Resource performance account 1 - SUT	ANR.EGR:BMK01
Resource performance account 2 - DCI	ANR.EGR:BMK02
Resource performance account 3 - LCI	ANR.EGR:BMK03
Resource performance account 4 - SCI	ANR.EGR:BMK04
Resource performance account 5 - IMN	ANR.EGR:BMK05
Resource performance account 6 - IMS	ANR.EGR:BMK06
Resource performance account 7 - SET	ANR.EGR:BMK07
Resource performance account 8 - STA	ANR.EGR:BMK08
Resource performance account 9 - U8	ANR.EGR:BMK09
Resource performance account 10 - U9	ANR.EGR:BMK10
Resource performance account 11 - MUT	ANR.EGR:BMK11
Resource performance account 12 - BKS	ANR.EGR:BMK12
Yield: base quantity unit	ANR.EGR:GUTB
Yield: primary quantity unit	ANR.EGR:GUTP / ANR.EGR:GUT
Yield: secondary quantity unit	ANR.EGR:GUTS
Yield: tertiary quantity unit	ANR.EGR:GUTT
Scrap: base quantity unit	ANR.EGR:AUSB
Scrap: primary quantity unit	ANR.EGR:AUSP / ANR.EGR:AUS
Scrap: secondary quantity unit	ANR.EGR:AUSS
Scrap: tertiary quantity unit	ANR.EGR:AUST

Actual values at the operation (OP status)

Rework quantity: base quantity unit	ANR.EGR:NCHB
Rework quantity: primary quantity unit	ANR.EGR:NCHP / ANR.EGR:NCH
Rework quantity: secondary quantity unit	ANR.EGR:NCHS
Rework quantity: tertiary quantity unit	ANR.EGR:NCHT
Problem quantity: base quantity unit	ANR.EGR:PRBB
Problem quantity: primary quantity unit	ANR.EGR:PRBP / ANR.EGR:PRB
Problem quantity: secondary quantity unit	ANR.EGR:PRBS
Problem quantity: tertiary quantity unit	ANR.EGR:PRBT
Actual cycle	ANR.IZY



We recommend defining the formulas in a template to allow for them to be assigned automatically (e.g. formula for remaining run time).



Deployment of MDE (only relevant if HYDRA is in use)

Different KPIs are also available in the MDE in the different evaluations. Refer to the corresponding documentation for the application to find out how the standard key figure is calculated and the name of the formula. The acronyms listed below are available for the following KPIs:

- Availability (avail)
- Quality (qual)
- NEE (nee)
- Planned production time (op_ti)
- Machine runtime (mch_rt)
- Yield utilization (yie_ut)
- Actual utilization (act_ut)
- Utilization efficiency (rcu)
- Allocation efficiency (ocu)
- Techn. Efficiency (tec_ef)
- Quote (yie_ra)
- Scrap ratio (scr_ra)

The following acronyms are available to define specific calculations. In the formula, use the acronym before "="; the acronym behind "=" is the service acronym used in the calculation (e.g. in the OEE report).

```

rpa1=efficiencyreport.rpa1
rpa2=efficiencyreport.rpa2
rpa3=efficiencyreport.rpa3
rpa4=efficiencyreport.rpa4
rpa5=efficiencyreport.rpa5
rpa6=efficiencyreport.rpa6
rpa7=efficiencyreport.rpa7
rpa8=efficiencyreport.rpa8
rpa9=efficiencyreport.rpa9
rpa10=efficiencyreport.rpa10
rpa11=efficiencyreport.rpa11
rpa12=efficiencyreport.rpa12
yield.primary=efficiencyreport.yield.primary
scrap.primary=efficiencyreport.scrap.primary
rework.primary=efficiencyreport.rework.primary
problem.primary=efficiencyreport.problem.primary
cycle_target_weighted=efficiencyreport.cycle_target_weighted
cycle_actual_weighted=efficiencyreport.cycle_actual_weighted
perf_rate_cycle_target=efficiencyreport.cycle_target_to_use
perf_rate_strokes=efficiencyreport.strokes_to_use
quality=efficiencyreport.quality
strokes=efficiencyreport.strokes
strokes.calc=efficiencyreport.strokes.calc
calculate_duration=efficiencyreport.calculate_duration
performance_rate → Place holder to ensure that the defined efficiency is always used. Available for the
KPI NEE and OEE.

```